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Continuous Gravitational Waves from Gravitars in Quadratic $F(R)$ Modified Gravity

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ABSTRACT

The paper delves into the continuous gravitational waves emitted from gravitars, neutron stars with no magnetic fields. Gravitational wave behavior is explored and compared in Einstein's general relativity and a quadratic $F(R)$ modified gravity theory. The intrinsic strains, gravitational wave frequencies, and detectabilities of gravitars are investigated, taking into account the Cosmic Explorer, Einstein Telescope, and aLIGO detectors. Employing a Monte Carlo Simulation of the galactic population, probability density functions for spatial positions, initial rotation periods, and ellipticity are utilized. Strains and frequencies within the galactic gravitar population are observed to follow a log-normal distribution. In addition, the maximum and average ellipticity values are determined to be near 10^{-6} and 10^{-7}. Furthermore, in quadratic $F(R)$ gravity, the galactic population's strain and frequency distributions is noted to be dependent upon the Ricci curvature R.

Keywords: Continuous Gravitational Waves, Isolated Neutron Stars, Modified Gravity Theories, $F(R)$ Gravity, Monte Carlo Simulation

1. INTRODUCTION

German physicist Albert Einstein formulated the theory of general relativity (GR) in 1915. A year later, he predicted the existence of gravitational waves (GWs), which are described as ripples in spacetime (Einstein 1916, 1918; Cervantes-Cota et al. 2016). In 1974, Alan R. Hulse and Joseph H.

Taylor discovered a binary pulsar (now known as the Hulse–Taylor pulsar or PSR B1913+16) that was losing orbital energy in agreement with GR's prediction of energy loss due to GWs. This discovery provided the first indirect evidence of the existence of GWs (Taylor et al. 1979; Cervantes-Cota et al. 2016). After a prolonged wait, on 14 September 2015, the Laser Interferometer Gravitational-Wave Observatory (LIGO) Hanford detector directly detected the first GW signal (known as GW150914) from a binary black hole coalescence, ultimately proving Einstein's prediction (Abbott et al. 2016; Cervantes-Cota et al. 2016).

GWs are classified based on their astronomical source, frequency, or behavior (i.e., duration and strength). This paper focuses on behavior-based classifications, namely: (1) continuous gravitational waves (CGWs); (2) inspirals (i.e., mergers) of compact binary systems; (3) burst signals like those from supernovae; and (4) stochastic background (e.g., from the early evolution of the universe) (Sieniawska & Bejger 2019). As of the time of writing, only inspiral GWs have been detected; however, this situation might change with LIGO's more sensitive fourth observing run (O4) currently underway (LIGO 2023). The primary emphasis of this paper lies in the realm of CGWs from neutron stars, particularly discussing CGWs arising from gravitars. Gravitars are rotating isolated neutron stars characterized by null magnetic fields, meaning that their radiation occurs exclusively through GWs instead of electromagnetic waves. Gravitars' CGWs can be emitted through asymmetrical deformations named mountains or oscillation modes (Haskell & Schwenzer 2021). This paper deals with deformation-driven CGWs only.

Even though GR has survived virtually all tests, alternative modified gravity (MG) theories are still worth consideration for multiple reasons: GR (1) does not unite the four fundamental forces, (2) requires adding the unexplained, unobserved dark sector (dark matter and dark energy), and (3) allows black hole singularities at which physics laws break (Shankaranarayanan & Johnson 2022). MG theories are often classified into four groups: (1) higher (than II) derivatives, which include the well-known $F(R)$ theories; (2) higher (than four) dimensions; (3) extra field contents; and (4) non-minimal coupling (Shankaranarayanan & Johnson 2022). The paper's main goal is to compare the behavior of gravitars' CGWs as described by GR and $F(R)$ MG theories. Specifically, the intrinsic

strain amplitude, spin-down rate, and detectability of gravitars are compared. This comparison is done through theoretical calculations and a Monte Carlo Simulation. The simulation utilizes probability density functions for a population of gravitars spatial positions, initial rotation periods, and ellipticity.

1.1. *Notation*

The space-like metric tensor signature (-,+,+,+) is used. Newton's notation for differentiation is used (i.e., $\dot{y} = \frac{dy}{dx}$, where y and x are variables). I do not use a Geometrized unit system ($c = G = 1$); instead, I use $G = 6.6743 \times 10^{-11}$ N m^2/kg^2 for Newton's gravitational constant and $c = 299,792,458$ m/s for the speed of light in vacuum. Additionally, I use $\exp(x)$ in place of e^x. Finally, I place symbols atop variables: straight lines denote the arithmetic mean, while tildes indicate an association with $F(R)$ gravity.

2. LITERATURE REVIEW

2.1. *Rotating Deformed Neutron Stars*

Given their compactness and extreme density, one might expect neutron stars to be perfectly spherical and symmetrical. However, neutron stars can, in fact, be asymmetric. For a rotating gravitar, its quadrupole deformations (sometimes called mountains) would cause CGWs emission. Such deformations can be due to the star's magnetic field or crust's elastic strain (Glampedakis & Gualtieri 2018). Since I intend to focus on gravitars with $B = 0$ T, I will consider only the latter cause. As shown in Figure 1, Gravitars can be described as rigidly rotating aligned (i.e., inclination angle $\tau = 0$) triaxial ellipsoids, with three moments of inertia along the three principal axes (Sieniawska & Bejger 2019).

Basing my approach on Neunzert (2019), I shall now derive two equations that I will use in the methods: the intrinsic strain amplitude h_0 function of distance d; and the GW frequency f_{GW} function of time t. Gravitars' deformation is measured by its (very small) quadrupole ellipticity:

$$\varepsilon = \frac{|I_{xx} - I_{yy}|}{I_{zz}}, \tag{2.1.1}$$

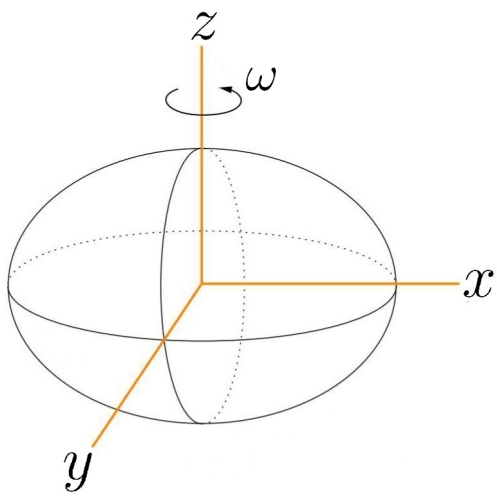


Figure 1: A non-axisymmetric gravitar (described as a triaxial ellipsoid) rotating with angular velocity ω about the z-axis. Edited from Neunzert (2019).

where I_{xx}, I_{yy}, and I_{zz} are the three principal moments of inertia along the x, y, and z axes. It is important to note that the z-axis is aligned with the spin axis. In this model, gravitars emit GWs at twice the spin frequency v:

$$f_{\mathrm{GW}} = 2v, \qquad v = \frac{\omega}{2\pi}, \tag{2.1.2}$$

where $\omega = \frac{2\pi}{P}$ is the angular velocity (sometimes called angular frequency) and P is the rotation period (sometimes called spin period). The quadrupole tensor of a gravitar is

$$I_0 = \begin{bmatrix} I_{xx} & 0 & 0 \\ 0 & I_{yy} & 0 \\ 0 & 0 & I_{zz} \end{bmatrix}. \tag{2.1.3}$$

Let the star be rotated by angle ωt in time t. Therefore, the rotation matrix function of time, $R_z(t)$, becomes

$$R_z(t) = \begin{bmatrix} \cos(\omega t) & \sin(\omega t) & 0 \\ -\sin(\omega t) & \cos(\omega t) & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{2.1.4}$$

The quadrupole tensor at time t becomes

$$I(t) = R_z(t) \times I_0 \times R_z^{-1}(t) \tag{2.1.5}$$

$$= \begin{bmatrix} \frac{1}{2}(I_{xx}+I_{yy}) + \frac{1}{2}(I_{xx}-I_{yy})\cos(2\omega t) & -\frac{1}{2}(I_{xx}-I_{yy})\sin(2\omega t) & 0 \\ -\frac{1}{2}(I_{xx}-I_{yy})\sin(2\omega t) & \frac{1}{2}(I_{xx}+I_{yy}) - \frac{1}{2}(I_{xx}-I_{yy})\cos(2\omega t) & 0 \\ 0 & 0 & I_{zz} \end{bmatrix}. \tag{2.1.6}$$

Differentiating with respect to time t twice, $\ddot{I}(t)$ is found to be

$$\ddot{I}(t) = 2\omega^2(I_{xx}-I_{yy}) \begin{bmatrix} -\cos(2\omega t) & \sin(2\omega t) & 0 \\ \sin(2\omega t) & \cos(2\omega t) & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{2.1.7}$$

Rewriting in terms of the ellipticity,

$$\ddot{I}(t) = 2\omega^2 I_{zz}\varepsilon \begin{bmatrix} -\cos(2\omega t) & \sin(2\omega t) & 0 \\ \sin(2\omega t) & \cos(2\omega t) & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{2.1.8}$$

2.2. *Einstein's General Relativity*

Deformations can be described independently from gravity theories, while the behavior of GWs is dependent upon gravity. Therefore, to reach the two desired equations, $h_0(d)$ and $f_{\mathrm{GW}}(t)$, choosing a gravity theory is necessary. For GR, the quadrupole formula is

$$h_{ij}^{TT} = \frac{2G}{c^4 d}\ddot{I}_{ij}, \tag{2.2.1}$$

where d is the gravitar-observer distance along the z-axis. I can then write

$$h_{ij}^{TT} = \frac{4G\varepsilon I_{zz}\omega^2}{c^4 r} \begin{bmatrix} -\cos(2\omega t) & \sin(2\omega t) & 0 \\ \sin(2\omega t) & \cos(2\omega t) & 0 \\ 0 & 0 & 1 \end{bmatrix}, \tag{2.2.2}$$

$$h_+ = -\frac{4G\varepsilon I_{zz}\omega^2}{c^4 r}\cos(2\omega t), \tag{2.2.3}$$

$$h_\times = \frac{4G\varepsilon I_{zz}\omega^2}{c^4 r}\sin(2\omega t). \tag{2.2.4}$$

Hence, the intrinsic strain amplitude h_0 is found to be

$$h_0(d) = \frac{4G\varepsilon I_{zz}\omega^2}{c^4 d} = \frac{4\pi^2 G\varepsilon I_{zz} f_{\rm GW}^2}{c^4 d}. \tag{2.2.5}$$

Since the torque $I_{zz}\dot{\omega}$ is the rate of change of momentum $\dot{J}_{zz} = -\frac{32G\varepsilon^2 I_{zz}^2\omega^2}{5c^2}$,

$$\dot{\omega} = -\frac{32G\varepsilon^2 I_{zz}\omega^5}{5c^5} \implies \dot{f}_{\rm GW} = -\frac{32\pi^4 G\varepsilon^2 I_{zz} f_{\rm GW}^5}{5c^5}. \tag{2.2.6}$$

which is are differential equations with the solutions

$$f_{\rm GW}(t) = \left(f_{{\rm GW},i}^{-4} + \frac{128\pi^4 G\varepsilon^2 I_{zz}}{5c^5}t\right)^{-\frac{1}{4}}, \qquad \dot{f}_{\rm GW}(t) = -\frac{1}{8\pi}\sqrt[4]{\frac{5c^5}{8G\varepsilon^2 I_{zz}t^5}}, \tag{2.2.7}$$

where $f_{{\rm GW},i}$ is the initial gravitational wave frequency (at $t = 0$). It is useful to rewrite $f_{{\rm GW},i}$ in terms of the initial rotation period P_i:

$$f_{\rm GW}(t) = \left(\frac{P_i^4}{16} + \frac{128\pi^4 G\varepsilon^2 I_{zz}}{5c^5}t\right)^{-\frac{1}{4}}. \tag{2.2.8}$$

2.3. *F(R) Modified Gravity*

$F(R)$ gravity is the most famous among all MG theories. It started in 1918, three years after Einstein published his GR theory (Weyl 1918; Clifton et al. 2012). $F(R)$ gravity generalizes GR by generalizing the Einstein–Hilbert action to a generic non-linear function of the Ricci curvature scalar R, which results in fourth-order field equations (Clifton et al. 2012). $F(R)$ gravity generalizes GR's Einstein–Hilbert action from

$$S_{\rm EH} = \frac{1}{2\kappa}\int d^4x\sqrt{-g}R + S_{\rm matter} \tag{2.3.1}$$

to

$$S_{F(R)} = \frac{1}{2\kappa}\int d^4x\sqrt{-g}F(R) + S_{\rm matter} \tag{2.3.2}$$

where $\kappa = \frac{8\pi G}{c^4}$ is Einstein gravitational constant, d^4x is the infinitesimal volume element in 4D spacetime, $g = \det(g_{\mu v})$ is the determinant of the metric tensor matrix, and $S_{\rm matter}$ is the action of the standard perfect fluid matter (Astashenok et al. 2013).

Each $F(R)$ function differs from one another. For example, GR's function is $F(R)_{\rm GR} = R$; as a result, every other $F(R)$ function can be expressed as

$$F(R) = R + f(R). \tag{2.3.3}$$

where $f(R)$ expresses the difference from GR gravity . Since $F(R)$ is infinitely differentiable, it can be represented by a Taylor series:

$$F(R) = \sum_{i=0}^{\infty} a_i R^i, \qquad f(R) = \sum_{i=2}^{\infty} a_i R^i. \tag{2.3.4}$$

where a_i is the ith coefficient.

For $F(R)$ theories, GWs behave differently because the effective gravitational constant $\tilde{G}$ ($\neq G$) is different; mathematically,

$$\tilde{G} = \frac{G}{\dot{F}(R)}, \tag{2.3.5}$$

where $\dot{F}(R) = 1 + \dot{f}(R)$ is the first derivative of $F(R)$ with respect to R (Inagaki & Taniguchi 2023). Therefore, the two equations, $h_0(d)$ and $f_{\rm GW}(t)$, for $F(R)$ gravity become

$$\tilde{h}_0(d) = \frac{4\pi^2 \tilde{G} \varepsilon I_{zz} \tilde{f}_{\rm GW}^2}{c^4 d}, \tag{2.3.6}$$

and

$$\tilde{f}_{\rm GW}(t) = \left(\frac{P_i^4}{16} + \frac{128\pi^4 \tilde{G} \varepsilon^2 I_{zz}}{5c^5} t \right)^{-\frac{1}{4}}. \tag{2.3.7}$$

2.4. *Monte Carlo Simulations*

A Monte Carlo Simulation (MCS) is a computational technique that uses random sampling, statistical modeling, and many experiments to reach conclusions about complex systems (Harrison 2010). MCSs usually make use of pseudo-random (computer-generated, i.e., not truly random) number generators and probability density functions (PDFs), which give the probability of a random variable falling within a particular range of values.

An MCS of a population of galactic neutron stars was first done by Palomba (2005); they simulated gravitars using PDFs for (exponential) radial displacement, (exponential) vertical displacement, kick velocity, (log-normal or uniform) initial rotation period, (uniform) age, (exponential) ellipticity, and galactic potential energy to estimate the CGWs' detectability by the first (O1) and second (O2) observing runs of the Virgo detector. Since then, multiple similar MCSs have been done for alike or different purposes (Sartore et al. 2010; Taani et al. 2012; Reed et al. 2021). Additionally, researchers

discovered more accurate PDFs for galactic neutron stars' variables as time went by. Radial displacement is best modeled by a gamma distribution (Sartore et al. 2010); vertical displacement is best modeled by an exponential distribution (Wei et al. 2010; Reed et al. 2021); and initial rotation period is best modeled by a log-normal distribution (Igoshev et al. 2022).

3. METHODS

To compare GR and $F(R)$ gravity, $F(R)$'s effective gravitational constant $\tilde{G}$ needs to be found using Equation 2.3.5. Therefore, finding $\dot{F}(R)$ requires the function $F(R)$ itself and the Ricci curvature R. $F(R)$ is just a matter of choice. R, on the other hand, needs to be determined. In fact, the Ricci curvature R outside a neutron star can be described as a zero-approaching oscillatory function of d, the distance from the neutron star center; however, complex considerations, such as neutron stars' Equation of State (EoS) models and the Tolman–Oppenheimer–Volkoff (TOV) equations, are required to do so (Astashenok et al. 2013; Resco et al. 2016). For the sake of simplicity, a set of positive and negative R values is chosen for comparison. The positive values ($10^{-3} \leq R \leq 10^{-7}/\mathrm{m}^2$) change logarithmically, while the negative values ($-10^{-6} \leq R < 10^{-5}$) are determined based on the positive values in such a way that they result in a multiplicative inverse effect on $\tilde{G}$.

I use an MCS to simulate a population of galactic gravitars. The number of galactic gravitars in the simulation is

$$N = \tau_{\mathrm{Birthrate}} \times \mathrm{Age}_{\mathrm{max}} = 10^6, \tag{3.1}$$

where $\tau_{\mathrm{Birthrate}} = 1/\mathrm{century}$ is the birthrate of neutron stars, and $\mathrm{Age}_{\mathrm{max}} = 100$ Myr is the maximum age of a (gravitationally or electromagnetically) detectable neutron star (Palomba 2005; Diehl et al. 2006; Rozwadowska et al. 2021). I use the canonical value for the moment of inertia along the z-axis I_{zz} of 10^{38} kg m^2.

Using a galactocentric cylindrical coordinate system shown in Figure 2, the simulation considers six variables' PDFs: (1) radial displacement $r = \sqrt{x^2 + y^2}$, (2) vertical displacement z, (3) azimuthal angle ϕ, (4) ellipticity ε, (5) initial rotation period P_i, and (6) age. This means that the inclination

angle ι, kick velocity v_{kick}, galactic potential energy Φ, and galactic spiral arms are effectively ignored for simplicity; however, this ignoration may reduce the accuracy of the results.

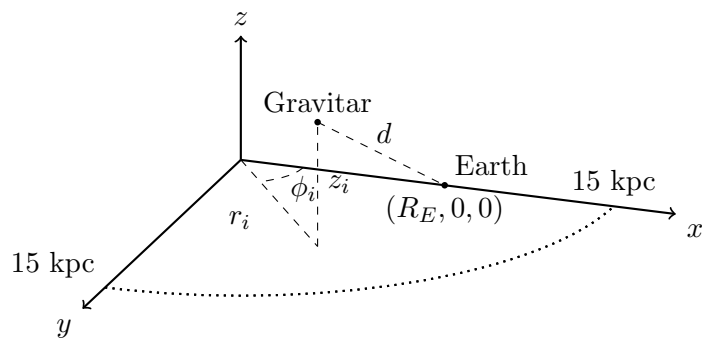


Figure 2: An arbitrary gravitar's galactocentric cylindrical coordinates (r_i, ϕ_i, z_i) shown in a three-dimensional Cartesian coordinate system, with Earth situated at a distance of d and positioned at coordinates $(R_E, 0, 0)$.

To generate a population of galactic gravitars that achieve the PDFs, I employ pseudo-random number generators (PRNGs) to generate random probabilities and find the variables at those given probabilities. However, PDFs (1) are not necessarily one-to-one functions, (2) don't necessarily have inverse functions, and (3) can vary in their output ranges. In order to address these considerations, I integrate PDFs to derive their cumulative distribution functions (CDFs). Mathematically, a CDF $P(x)$ can be expressed as

$$P(x) = \int_{x_{\min}}^{x} p(x)\ dx, \qquad x_{\min} \leq x \leq x_{\max}, \tag{3.2}$$

where $x_{\min}$ is x's minimum value, $x_{\max}$ is x's maximum value, $p(x)$ is x's PDF, and $P(x)$ gives the cumulative probability of being less than or equal to x (Harrison 2010). Since normalized PDFs, which exhibit a total probability of 1, inherently possess invertible CDFs that span an output range from 0 to 1, I can pair PRNGs with CDFs' inverses to generate the desired population of galactic neutron stars.

Adapted from Van der Kruit (1987), Paczynski (1990), Sartore et al. (2010), Wei et al. (2010), and Taani et al. (2012), the radial displacement r distribution is given by the gamma function $\Gamma(2, 4.5)$:

$$p(r) = a_r \frac{r}{R_{\text{exp}}^2} \exp\left[-\frac{r}{R_{\text{exp}}}\right], \tag{3.3}$$

where

$$a_r = \left[1 - \left[1 + \frac{R_{\max}}{R_{\exp}}\right] \exp\left[-\frac{R_{\max}}{R_{\exp}}\right]\right]^{-1} \approx 1.18285 \tag{3.4}$$

is the normalizing constant, $R_{\exp} = 4.5$ kpc is the peak location, and $R_{\max} = 15$ kpc is the maximum radial displacement, because neutron stars hardly form beyond the stellar disk radius of 15 kpc (Jones & Robert 2004; Taani et al. 2012).

Adapted from Gott III et al. (1970), Gonthier et al. (2002), Palomba (2005), Wei et al. (2010), and Taani et al. (2012), the vertical displacement z distribution is given by an exponential function:

$$p(z) = \frac{1}{h_z} \exp\left[-\frac{|z|}{h_z}\right], \tag{3.5}$$

where $h_z = 0.075$ kpc is the scale height. Previous literature presented the distribution as $p(z) = \frac{1}{2h_z} \exp\left[-\frac{|z|}{h_z}\right]$ when z ranged from $-\infty$ to ∞. As the distribution is symmetric about 0 due to the absolute function, I have omitted the factor of half and am considering z within the range of 0 to ∞.

The azimuthal angle ϕ distribution is distributed uniformly from 0 to π, where

$$x = r\cos(\phi), \qquad y = r\sin(\phi). \tag{3.6}$$

As shown in Figure 2, Earth's coordinates are $x = R_E$, $y = 0$, $z = 0$, and (consequently) $\phi = 0$. The Earth-gravitar distance, therefore, becomes

$$d = \sqrt{(r\cos(\phi) - R_E)^2 + (r\sin(\phi))^2 + z^2} \tag{3.7}$$

where $R_E = 8.2$ kpc is the distance from the galactic center to the Earth (Abuter et al. 2019).

Adapted from Palomba (2005), the ellipticity ε distribution is given by an exponential function:

$$p(\varepsilon) = \frac{a_\varepsilon \exp\left[-\frac{\varepsilon}{\overline{\varepsilon}}\right]}{\overline{\varepsilon}\left(1 - \exp\left[-\frac{\varepsilon_{\max}}{\overline{\varepsilon}}\right]\right)}, \tag{3.8}$$

where

$$a_\varepsilon = \left[\frac{\exp\left[-\frac{\varepsilon_{\min}}{\overline{\varepsilon}}\right] - \exp\left[-\frac{\varepsilon_{\max}}{\overline{\varepsilon}}\right]}{1 - \exp\left[-\frac{\varepsilon_{\max}}{\overline{\varepsilon}}\right]}\right]^{-1} \tag{3.9}$$

is the normalizing constant (which is needed to account for the minimum ellipticity $\varepsilon_{\min} = 10^{-9}$), $\varepsilon_{\max}$ is the maximum ellipticity, and $\overline{\varepsilon}$ is the average ellipticity[1] (Woan et al. 2018). I am varying both

[1] In Palomba's paper, they denoted this with τ. Using their provided equation, $\overline{\varepsilon} = \tau - \frac{\varepsilon_{\max}}{\exp\left[\frac{\varepsilon_{\max}}{\tau}\right]-1}$, τ can be numerically equated to $\overline{\varepsilon}$.

the maximum and average ellipticities. The maximum ellipticity $\varepsilon_{\max}$ will be either 10^{-6} or 10^{-5}, encompassing the value obtained by Morales & Horowitz (2022). The average ellipticity $\bar{\varepsilon}$ will take on values of either 10^{-7} (as per Talukder et al. (2014)) or 10^{-6}. For low ellipticities (i.e., $\varepsilon_{\max} = 10^{-6}$ and $\bar{\varepsilon} = 10^{-7}$), the normalizing constant is $a_\varepsilon \approx 1.01005$. Conversely, for high ellipticities (i.e., $\varepsilon_{\max} = 10^{-5}$ and $\bar{\varepsilon} = 10^{-6}$), the normalizing constant is $a_\varepsilon \approx 1.00100$.

Adapted from Igoshev et al. (2022), the initial rotation period P_i distribution is given by a 10-base log-normal function:

$$p(P_i) = \frac{\log_{10}(e)}{P_i\sqrt{2\pi}\sigma_P} \exp\left(-\frac{(\log_{10}(P_i) - \overline{\log_{10}(P_i)})^2}{2\sigma_P^2}\right), \tag{3.10}$$

where $\overline{\log_{10}(P_i)} = -1.04$ and $\sigma_P = 0.53$ are the mean and standard deviation of $\log_{10}(P_i)$. Finally, the age is distributed uniformly from 0 to 100 Myr.

Defining PRNG(x) as a PRNG that selects a random value from 0 to x, the simulation steps for GR can be summarized in Figure 3. It is important to note that the PDFs are not used directly; instead, their invertible CDFs are found using Eq. 3.2 and then used, as discussed previously.

Each gravitar's detectability (which is a Boolean value) is determined depending on its f_{GW} and h_0 values compared to an interferometer GW detector's sensitivity curve. Specifically, the (O3) Advanced LIGO (aLIGO), Einstein Telescope, and Cosmic Explorer detectors' sensitivity curves are considered.

For $F(R)$ gravity, I calculate $\dot{F}(R)$, select an R value, use Equation 2.3.5 to compute $\tilde{G}$, and follow a similar procedure as the one shown in Figure 3; instead of utilizing Eq. 2.2.8 and Eq. 2.2.5, I replace them with Eq. 2.3.7 and Eq. 2.3.6 to account for the different gravitational constant. The chosen $F(R)$ theory is quadratic and is described by

$$F(R) = R + \alpha R^2, \qquad \dot{F}(R) = 1 + 2\alpha R, \tag{3.11}$$

where $|\alpha| = 10^5$ m^2 (Arapoğlu et al. 2011). The choice of a negative α has a specific implication. To maintain $\tilde{G} > 0$ with a negative α, the value of R must be less than 10^{-5}, which limits the effective range to $10^{-6} \leq R < 10^{-5}$ /m^2.

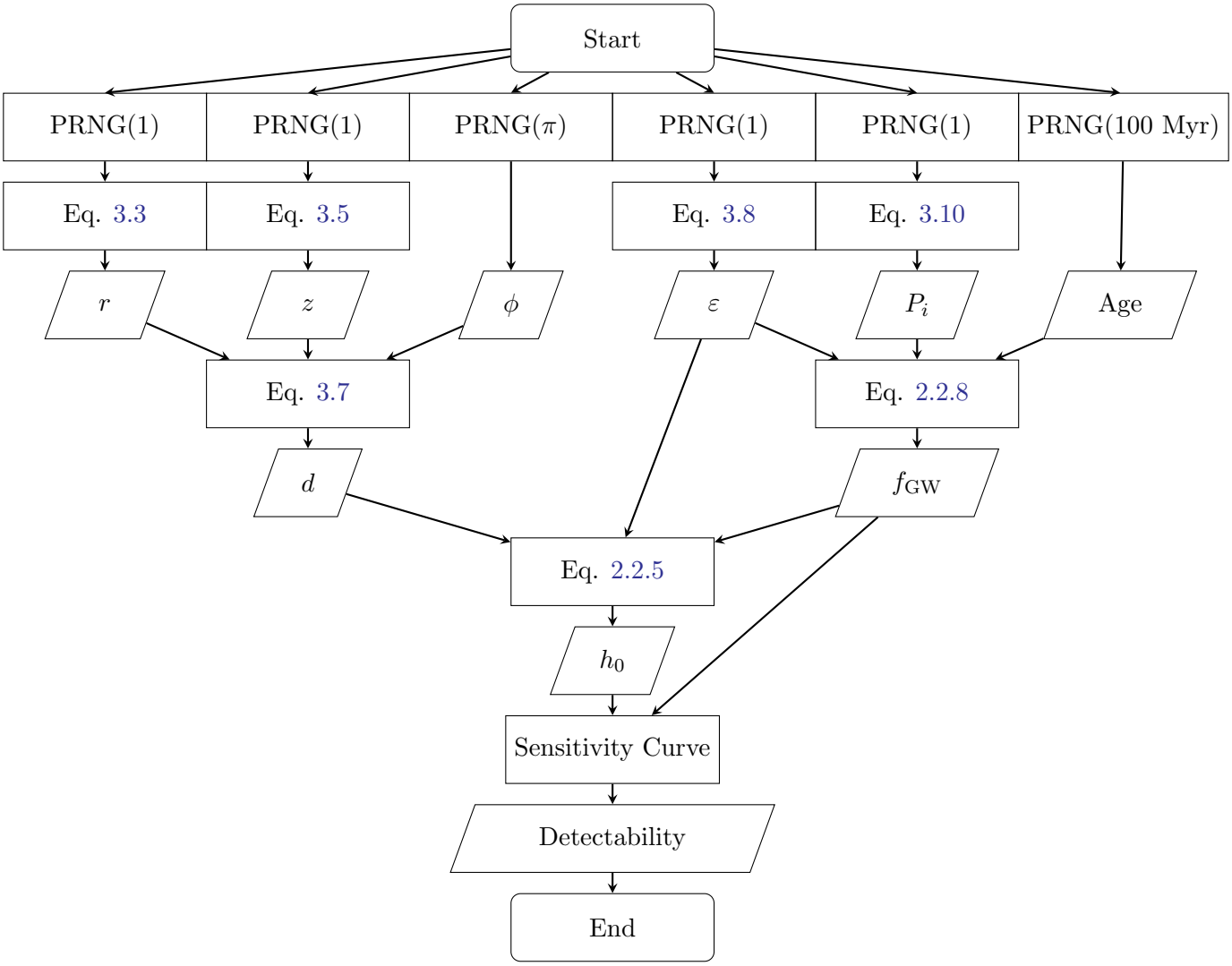


Figure 3: A flowchart depicting the Monte Carlo Simulation steps for general relativity. These steps are iterated $N = 10^6$ times, generating a population of galactic gravitars.

In contrast, for convention, opting for a positive α (i.e., $\alpha = 10^5$ m^2) allows for a broader range of positive R values. This choice effectively widens the range of effective positive R values ($10^{-5} < R$) while narrowing the range of effective negative R values ($-10^{-6} \leq R < -10^{-5}$ /m^2).

4. RESULTS AND DISCUSSION

4.1. *General Relativity*

The effective gravitational constant in GR remains unchanged because $\dot{F} = 1$. Therefore, the quadratic $F(R)$ theory simplifies to GR when $R = 0$. Investigating these results in GR allows for a

comparison with $F(R)$ gravity. Prior to comparing $F(R)$ with GR, populations with low and high ellipticities are compared in GR.

4.1.1. *Low Ellipticities*

For low ellipticities ($\varepsilon_{\max} = 10^{-6}$ and $\bar{\varepsilon} = 10^{-7}$), the intrinsic strains h_0 and gravitational wave frequencies $f_{\rm GW}$ are log-normally distributed, as shown in Figure 4. The distributions of the two variables are accurately modeled by

$$p(f_{\rm GW}) = \frac{1}{f_{\rm GW}\sigma_{f_{\rm GW}}\sqrt{2\pi}} \exp\left[-\frac{(\log_{10}(f_{\rm GW}) - \overline{\log_{10}(f_{\rm GW})})^2}{2\sigma_{f_{\rm GW}}^2}\right], \tag{4.1.1}$$

and

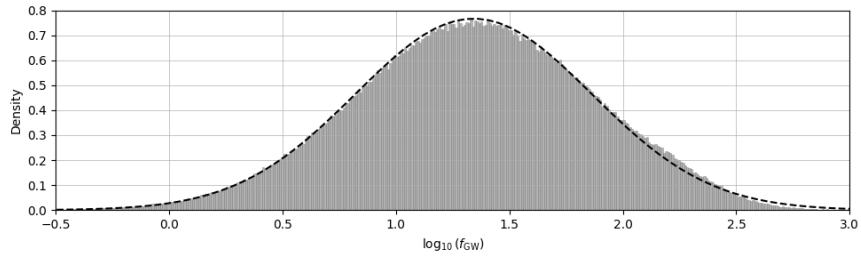
$$p(h_0) = \frac{1}{h_0\sigma_h\sqrt{2\pi}} \exp\left[-\frac{(\log_{10}(h_0) - \overline{\log_{10}(h_0)})^2}{2\sigma_h^2}\right], \tag{4.1.2}$$

where $\overline{\log_{10}(f_{\rm GW})} = 1.34$ and $\sigma_{f_{\rm GW}} = 0.52$ are the mean and standard deviation of $\log_{10}(f_{\rm GW})$, respectively. Similarly, $\overline{\log_{10}(h_0)} = -29.48$ and $\sigma_h = 1.18$ are the mean and standard deviation of $\log_{10}(h_0)$.

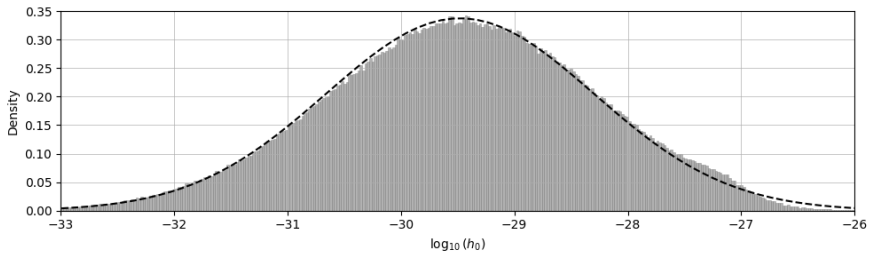
This finding is surprising because both h_0 and $f_{\rm GW}$ involve various operations on variables with different distributions (see Figure 3). Additionally, Figure 4 shows how galactic gravitars would be hardly detectable as virtually no stars attain an intrinsic strain of greater than 10^{-25}.

To gain more insight, I plot $\log_{10}(h_0)$ against $\log_{10}(f_{\rm GW})$ to compare the population of gravitars with the sensitivity curves of aLIGO, Einstein Telescope, and Cosmic Explorer using data from LIGO (2016). As shown in Figure 5, for low ellipticities, galactic gravitars' frequencies and strains would be concentrated between 5–100 Hz and 10^{-31}–10^{-28}. No gravitars exceed $h_0 = 10^{-25}$; therefore, none are detectable by any of the three detectors. However, some gravitars approach $h_0 = 10^{-25}$, specifically at frequencies of 100–1000 Hz. Additionally, when $\log_{10}(h_0)$ is plotted against $\log_{10}(f_{\rm GW})$, the population's minimum, average, and maximum $\log_{10}(h_0)$ seem to increase linearly with $\log_{10}(f_{\rm GW})$; in a surprisingly accurate fashion, the minimum appears to be described by

$$\log_{10}(h_0)_{\min} = 2\log_{10}(f_{\rm GW}) - 34.27. \tag{4.1.3}$$



(a) A 10-base log-normal distribution of f_{GW}. The mean and standard deviation of $\log_{10}(f_{\mathrm{GW}})$ are 1.34 and 0.52.



(b) A 10-base log-normal distribution of h_0. The mean and standard deviation of $\log_{10}(h_0)$ are -29.48 and 1.18.

Figure 4: 10-base log-normal distributions of h_0 and f_{GW} accurately fitting their 500-bin histograms when $\varepsilon_{\max} = 10^{-6}$ and $\overline{\varepsilon} = 10^{-7}$ in general relativity.

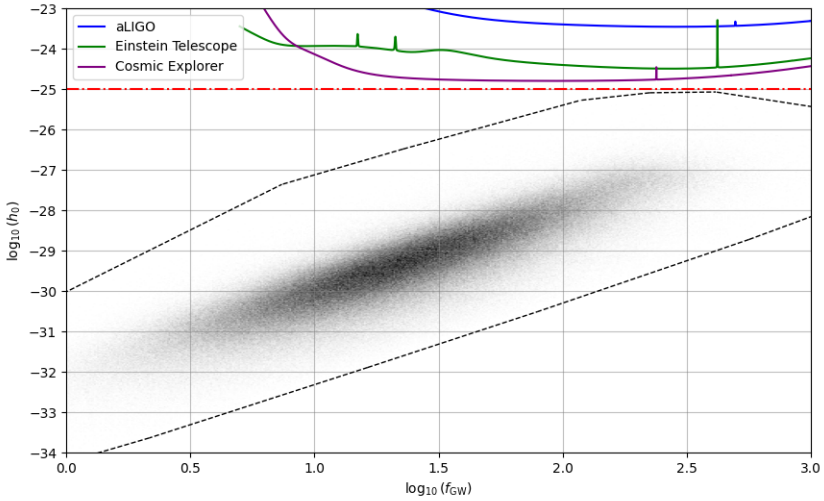
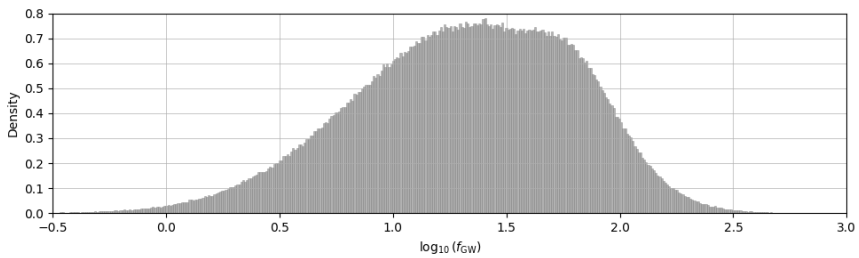


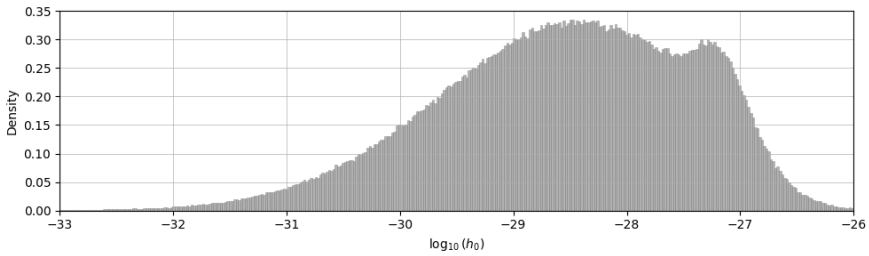
Figure 5: CGWs strain-frequency distribution compared with aLIGO, Einstein Telescope, and Cosmic Explorer detectors' sensitivity curves when $\varepsilon_{\max} = 10^{-6}$ and $\overline{\varepsilon} = 10^{-7}$ in general relativity. Detectors' data source is LIGO (2016).

4.1.2. *High Ellipticities*

For high ellipticities ($\varepsilon_{\max} = 10^{-5}$ and $\overline{\varepsilon} = 10^{-6}$), the intrinsic strains h_0 and gravitational wave frequencies f_{GW} have non-typical distributions (see Figure 6). As shown in Figure 6a, the mean and standard deviation of $\log_{10}(f_{\mathrm{GW}})$ decreased, and its distribution became broader at 10—100 Hz compared to the distribution for low ellipticities. On the other hand, Figure 6b shows that the mean $\log_{10}(h_0)$ increased, implying that the more gravitars have higher CGWs strains in higher ellipticities. There also seems to be an unusual density increase at $10^{-27.5}$–$10^{-26.5}$.



(a) A 500-bins histogram of $\log_{10}(f_{\mathrm{GW}})$. The mean and standard deviation are 1.32 and 0.48.



(b) A 500-bins histogram of $\log_{10}(h_0)$. The mean and standard deviation are −28.55 and 1.12.

Figure 6: Unusual distributions of h_0 and f_{GW} when $\varepsilon_{\max} = 10^{-5}$ and $\overline{\varepsilon} = 10^{-6}$ in general relativity.

Plotting $\log_{10}(h_0)$ against $\log_{10}(f_{\mathrm{GW}})$ reveals that some gravitars exceed $h_0 = 10^{-25}$ and are detectable by the Cosmic Explorer, as shown in Figure 7. However, for high ellipticities, these high GW signals occur at lower frequencies, precisely at 40–290 Hz, in contrast to lower ellipticities. Abbott et al. (2019) showed that CGWs strain upper limits shown in Figure 8 are very close to 10^{-25} at 50–1000 Hz, which contradicts the results for high ellipticities shown in Figure 7. In fact, Reed et al. (2021) demonstrated that CGWs suggest that less than 0.01% of galactic neutron stars have

ellipticities higher than 10^{-6}. Therefore, from this point onward, it can be assumed that the low ellipticities model with $\varepsilon_{\max} = 10^{-6}$ and $\overline{\varepsilon} = 10^{-7}$ is more accurate than the high ellipticities model with $\varepsilon_{\max} = 10^{-5}$ and $\overline{\varepsilon} = 10^{-6}$. Only the low ellipticities model will be used in $F(R)$ modified gravity.

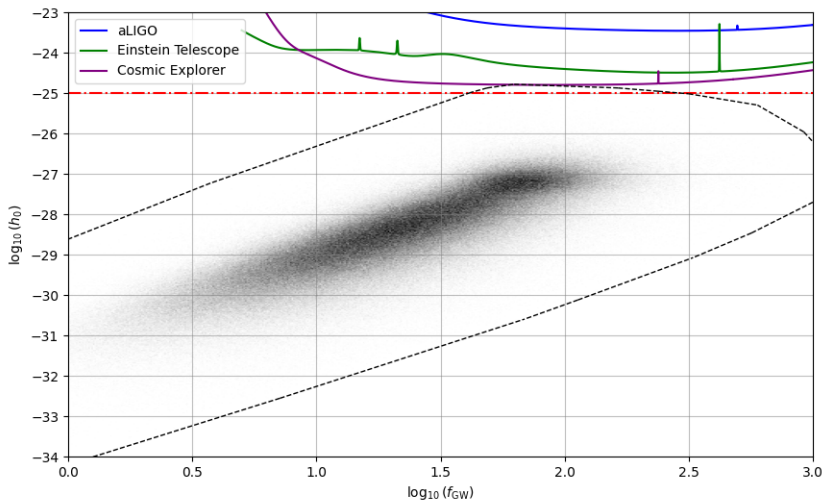


Figure 7: CGWs strain-frequency distribution compared with aLIGO, Einstein Telescope, and Cosmic Explorer detectors' sensitivity curves when $\varepsilon_{\max} = 10^{-5}$ and $\overline{\varepsilon} = 10^{-6}$ in general relativity. Detectors' data source is LIGO (2016).

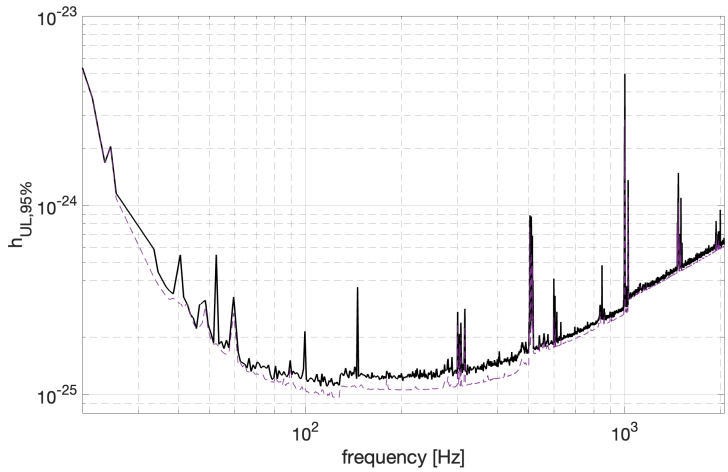
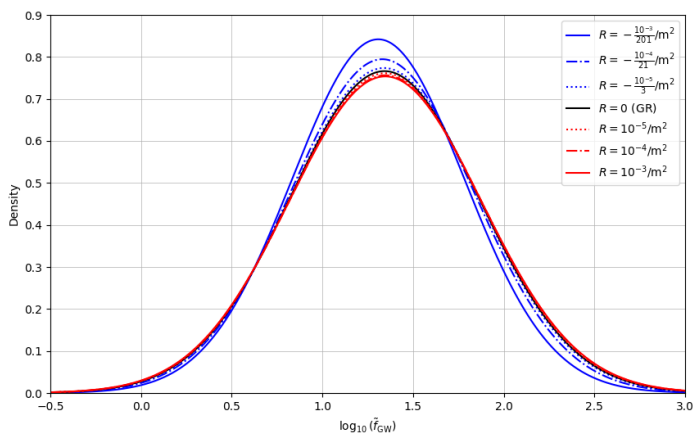


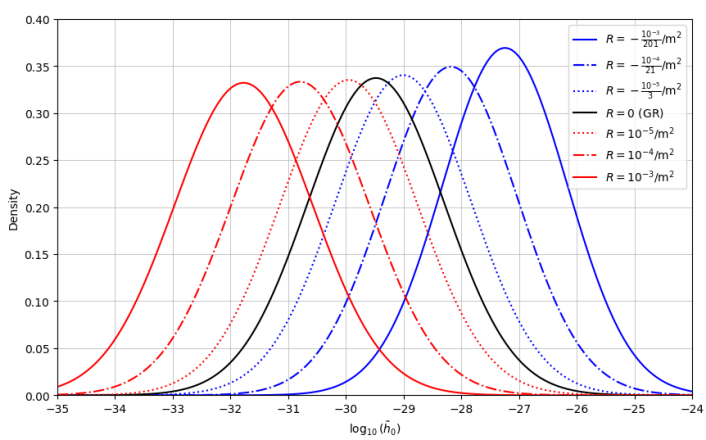
Figure 8: aLIGO's CGWs strains' upper limits (bold curve) and sensitivity lower bound (light dashed curve). Reprinted from Abbott et al. (2019).

4.2. *Quadratic $F(R)$ Modified Gravity*

In Subsubsection 4.1.2, I demonstrated that a model with lower ellipticities is accurate for the galactic population, unlike the model with higher ellipticities. Furthermore, in Subsubsection 4.1.1, I showed that the strains and frequencies of galactic gravitars with low ellipticities can be accurately modeled using log-normal distributions. With these insights in mind, I am using Eq. 4.1.1 and Eq. 4.1.2 to present the results of $F(R)$ gravity.



(a) Variations in populations' CGWs frequency distribution across a range of R values.



(b) Variations in populations' CGWs strain distribution across a range of R values.

Figure 9: Variations of CGWs strains and frequencies' 10-base log-normal distributions as R varies in quadratic $F(R)$ gravity with $\alpha = 10^5$ m^2.

Figure 9 illustrates the changes in the distributions of $\tilde{h}_0$ and $\tilde{f}_{\rm GW}$ resulting from variations in $\tilde{G}$ through variations in R (see Eq. 2.3.5 and Eq. 3.11). The distributions of $\tilde{f}_{\rm GW}$ exhibit minimal changes as R varies, especially when $R > 0$. In fact, when R increases from $-\frac{10^{-3}}{201}$ to 0, the mean and standard deviation of $\log_{10}(\tilde{f}_{\rm GW})$ increase by 2.6% and 9.9%, respectively. On the other hand, as R increases from $-\frac{10^{-3}}{201}$ to 0, the mean and standard deviation of $\log_{10}(\tilde{f}_{\rm GW})$ increase by only 0.3% and 1.6%, respectively.

The limited impact of positive R values on the $\log_{10}(\tilde{f}_{\rm GW}$ distribution can be partly attributed to the minor role that $\tilde{G}$ plays in influencing the value of $\tilde{f}_{\rm GW}$, as described in Eq. 2.3.7. Since the distributions of $\tilde{f}_{\rm GW}$ for populations with $R > 0$ values remain unchanged compared to populations with $R < 0$ values, I can conclude that for $R > 0$, it can be assumed that the population distribution remains unaffected, especially when R is near 0.

In contrast, Figure 9b shows that the distributions of $\tilde{h}_0$ change noticeably as $R > 0$ increases. The mean of $\log_{10}(\tilde{h}_0)$ decreases from -29.48 to -31.77 as R increases from 0 to 10^{-3}. However, the standard deviation of $\log_{10}(\tilde{h}_0)$ does not change as significantly: it increases from 1.18 to 1.20 as R increases from 0 to 10^{-3}. Values of $R \leq 10^{-6}/\mathrm{m}^2$ have insignificant effects, so they are not included in Figure 9b. The effects of $R = 10^{-5}/\mathrm{m}^2$, $R = 10^{-4}/\mathrm{m}^2$, and $R = 10^{-3}/\mathrm{m}^2$ can respectively be described as minimal, noticeable, and substantial.

Conversely, negative $R > 0$ values have a greater effect on the standard deviations of $\log_{10}(\tilde{h}_0)$; as R increases from from $-\frac{10^{-3}}{201}$ to 0, the standard deviation increases from 1.08 to 1.18. The effect on the means, however, is similar to that of positive R values; as R increases from $-\frac{10^{-3}}{201}$ to 0, the mean decreases from -27.24 to -29.48. As a general rule, increasing R results in the populations' $\log_{10}(\tilde{h}_0)$ standard deviations increasing (especially when $R < 0$) and $\log_{10}(\tilde{h}_0)$ means decreasing.

Since $\tilde{f}_{\rm GW}$ remains relatively constant as R increases, Eq. 2.3.6 is employed to visualize $\tilde{h}_0(R > 0)$ for various distances d, as shown in Figure 10. The values $\overline{\varepsilon} = 10^{-7}$ and $\overline{f}_{\rm GW} = 43.27$ Hz are set as constants (because they are the means). The chosen distances are (1) $d = 1$ kpc, (2) $d = 8.5$ kpc, representing the most probable distance, and (3) $d = 23.5$ kpc, representing the maximum distance. Furthermore, a curve connecting the average $\tilde{h}_0$ values for each population at specific R values is

plotted. It is evident that this line connecting the points closely, but not perfectly, resembles the curves generated using Eq. 2.3.6. The slight difference is suspected to be due to two factors: (1) the small size of the data (only five points) and (2) the approximation of treating $\tilde{f}_{\mathrm{GW}}$ as a constant rather than a function of R.

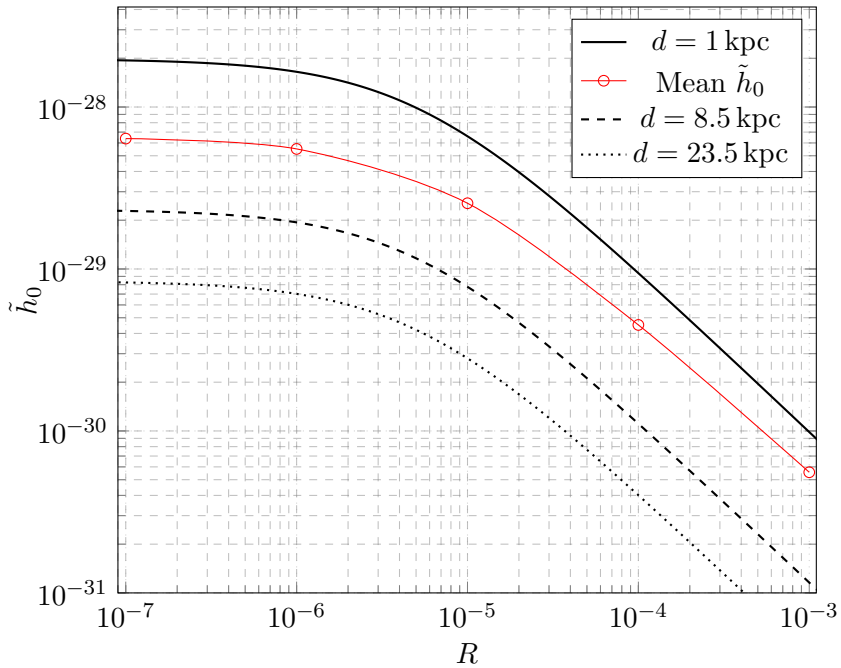


Figure 10: Variation of $\tilde{h}_0$ with $R > 0$ in quadratic $F(R)$ gravity. The red line connects the mean strains for different galactic populations, each associated with a different R value.

5. CONCLUSIONS

The study found that the CGW strains and frequencies within the galactic population of gravitars follow a log-normal distribution. In addition, it was determined that accurate values for $\varepsilon_{\mathrm{max}}$ and $\overline{\varepsilon}$ are close to 10^{-6} and 10^{-7}, respectively, aligning with previous literature and all-sky searches. These ellipticity values have led to the conclusion that, within the General Relativity (GR) framework, no galactic gravitars are detectable by the Cosmic Explorer, Einstien Telescope, or aLIGO detectors.

Nevertheless, it is noteworthy that some simulated gravitars displayed CGW strains reaching levels near 10^{-25} at frequencies higher than 100 Hz. This hints at the potential for detecting the CGW

signals, with Cosmic Explorer and the Einstein Telescope being the most sensitive detectors within the 100-1000 Hz range.

The study found effects on galactic populations' strain and frequency distributions in quadratic $F(R)$ gravity, with $\alpha = 10^5$ m^2. As R values increase, there is a notable increase in the mean and standard deviation of $\log_{10}(\tilde{f})_{\mathrm{GW}}$, particularly when $R < 0$. At the same time, standard deviations of $\log_{10}(\tilde{h}_0)$ increase, especially for $R < 0$, while $\log_{10}(\tilde{h}_0)$ decrease.

Interestingly, while $F(R)$ gravity exerts a measurable influence on gravitational wave frequency, its impact on CGW frequency remains relatively negligible for positive R. Therefore, the non-detection of CGWs could slightly support modified gravity.

To enhance the study's generality, the incorporation of probability density functions (PDFs) for variables such as inclination angle (ι), kick velocity (v_{kick}), galactic potential energy (Φ), and galactic spiral arms holds the potential to yield more robust results.

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APPENDIX

A. CONSTANTS AND BOUNDS

Table 1: The Monte Carlo Simulation constants.

Constant	Essence	Value	Source
a_r	Normalizing Constant	1.18285	Eq. 3.4
R_{exp}	Peak Location	4.5 kpc	Taani et al. (2012)
h_z	Scale Height	0.075 kpc	Taani et al. (2012)
R_E	Distance to the Galactic Center	8.2 kpc	Abuter et al. (2019)
a_ε	Normalizing Constant	1.010051	Eq. 3.9
$\overline{\varepsilon}$	Ellipticity Mean	10^{-7}	Talukder et al. (2014)
$\overline{\log_{10}(P_i)}$	Period's Logarithm Mean	-1.04	Igoshev et al. (2022)
σ_P	Period's Logarithm Standard Deviation	0.53	Igoshev et al. (2022)
I_{zz}	Moments of Inertia Along the z-axis	10^{38} kg m^2	–

Table 2: Variables' upper and lower limits.

Variable	Symbol	Minimum	Maximum
Radial Displacement	r	0	15 kpc
Vertical Displacement	z	0	∞
Azimuthal Angle	ϕ	0	π
Ellipticity	ε	10^{-9}	10^{-6}
Initial Period	P_i	0	∞
Age	–	0	100 Myr

B. GITHUB REPOSITORY

The Monte Carlo simulation codes for both GR and $F(R)$ theories are available on the GitHub repository. The repository also includes some figures and the corresponding codes used to generate them.

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